Fan XieYi (Ivan)

AI/Agent & Full-Stack Engineer

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Education

The University of Hong Kong, BEng (Sep 2023 - Jun 2027) in Computer Engineering

- 2023 Dean's Honour List

Skills

Programming: Python, TypeScript, Java, Cpp/C, PHP, SQL

AI/Agent: LangGraph/LangChain, PyTorch(CNN, RNN, LSTM, Transformer, VAE, VLA), pandas, scikit-learn, SciPy

RL: DQN, DQN Variants, GRPO, PPO, SAC

FullStack: FastAPI, express.js, Node.js, React, React Router, Tailwind CSS, Vite, Shadcn/ui, SQLite, MySQL, PostgreSQL, MongoDB

2023

Full-Stack Developer, TripleUni

Independently developed a service health monitoring system for the TripleUni three-university treehole platform (HKU, CUHK, HKUST; **20,000+** registered users). Aimed to provide a unified platform to check and manage the health of multiple services for fast anomaly fixing. Built with apschedule, FastAPI, and SQLite.

- **Monitoring Platform**: Built a proactive health-monitoring platform with HTTP/WebSocket/SSE probes covering 12+ core services; on service failure, **real-time Lark (Feishu) alerts** with service status and logs always visible — cutting bug discovery time from up to **72 hours** (waiting for user reports) to **minutes** (probe intervals: 1 minute for read APIs, 1 hour for write APIs)
- **Architecture Improvement**: Led the backend refactor, established a separate testing server, and standardized the CI/CD process, eliminating direct-to-production deployments and related incidents

2024

Full-Stack Developer, HKU Innovation Wing Lab

Built an AI system for Hong Kong's cultural heritage using historical archives from the HKU Library. Used MinerU and VLM to build a bilingual PDF preprocessing pipeline, and LangChain to build the RAG workflow.

- Addressed the open-source MinerU tool's failure to recognize image descriptions embedded in body paragraphs, which degraded multimodal-retrieval captions: collected image neighborhoods, 3–5 text blocks before/after on the same page, and cross-page candidates; filtered candidates using coreference/caption keywords, text-image distance vectors, and non-semantic caption IDs; applied VLM fine re-ranking when multiple candidates remained, and embedded the image link at the page end as a fallback when none did — improving image-matching accuracy from **40% to 100%**.
- Integrated Agent web search and designed a multi-source data filtering mechanism: to counter off-target information and hallucination risks from low-authority sources (Baidu Baike, Wikipedia, community blogs, etc.), established a whitelist of authoritative historical data websites, narrowing retrieval to trusted historical sources — eliminating non-authoritative interference and improving RAG response accuracy and credibility.

2025

Full-Stack Developer, HKU Innovation Wing Lab

An intelligent tour guide system developed in collaboration with the HKU UMAG Museum to address the shortage of docents. The app provides real-time voice interaction and multimodal information retrieval to help users better understand the artifacts in the collection. (LangGraph, ONNX, Agentic RAG)

- Addressed parsing corruption after MinerU processed bilingual (Chinese-English) image-text documents — mixed Chinese-English content and captions wrongly inserted into body text — by designing a language-separation program based on Chinese/English character ratios and rule-specific regular expressions; reordered sentences using Sentence Transformer sentence-order likelihood (P(A,B)>P(B,A) determines the plausible order); and corrected caption positions by judging text-image proximity via Euclidean distance, effectively improving document parsing quality.
- Designed and implemented a Reflection-RAG self-reflection mechanism to counter model hallucinations of names, places, and dates in high-precision historical narration: after generation, a reflection node evaluates whether the response resolves the user query and whether it contains hallucinations not grounded in the provided documents; correct responses are returned directly, while incorrect ones are sent back to the Generator for regeneration with feedback (unanswered parts, hallucinated parts); when a complete response is impossible due to missing document information, Web Search is triggered to supplement from the internet — improving response accuracy and completeness.
- Addressed keyword-retrieval failures caused by referring expressions and vague wording (e.g., "similar artifact") in multi-turn RAG conversations by introducing a Query Rewriter that rewrites references and vague terms into concrete entities from the previous turn, improving query specificity and keyword recall.
- Addressed generic visitor questions (e.g., "tell me about this artifact") that failed to retrieve deep information such as craftsmanship, historical background, and cultural significance by designing a Query Decomposition + Enhancement retrieval mechanism: first decompose the original query into multiple sub-queries to enrich the question, then use Q2D on each sub-query to generate pseudo-responses and concatenate them with the original query, introducing more context to improve keyword matching and relevant-document recall; the original query is also retained in retrieval to suppress pseudo-document hallucinations — improving information coverage and retrieval relevance for generic queries.
- **Multimodal Retrieval**: Built a Dino-based image vector retrieval engine enabling photo-to-artifact search; in a Dino vs CLIP comparison, Dino significantly outperformed CLIP in retrieval accuracy on cultural artifacts.
- Built a self-constructed 1K evaluation dataset to validate the optimal hybrid search (keyword + semantic + sparse) combination; reused AIHA image-text matching techniques with MinerU + DeepSeek-OCR to process **200+ pages** of bilingual Chinese-English PDFs, generating **5,000+ knowledge entries**.
- Deployed TTS and ASR models on-device with ONNX, reducing request-to-first-TTS-playback latency by **50%**.

Full-Stack Developer, HKUMed IT Department – Research Data Collaboration Catalogue

Built an internal research collaboration platform for the HKU Medical Faculty to facilitate cross-disciplinary collaboration among 200+ professors. (Next.js · FastAPI · PubMed API)

- Cleaned and standardized the profiles, publications, and collaboration data of **200+ professors**, and built the complete application with Next.js
- **Author name disambiguation**: resolved Chinese/English name variants and inconsistent author signatures across papers/journals using official website information, name normalization, and fuzzy search
- **Paper deduplication**: deduplicated papers across sources via SimHash + metadata matching; designed a medical keyword search engine (stemming, fuzzy matching, multi-dimensional filtering) and integrated the PubMed API for automatic paper crawling

2026

Research Assistant (Embodied Intelligence), HKU – Prof. YanChao Yang Research Group

Researching how to incorporate multimodal data (EEG, EMG, eye gaze) to enhance VLA model performance. (PyTorch, ALOHA, EEG/EMG)

- Proposed a VLA enhancement method that fuses multimodal data only during training and requires no multimodal input pairs at inference: collects three additional modalities — EEG, EMG, and eye gaze — embeds two generator modules in the OpenVLA architecture to simulate brain EEG/EMG signals, and aligns the generator distributions with real data distributions during training via KL divergence; uses the eye-gaze distribution for normalization in the last attention layer, before the action layer — enhancing VLA capability without adding inference-time multimodal pairing dependencies.

Projects

From-Scratch Byte-Level BPE Tokenizer & Small LLM Pre-training

- Implemented a byte-level BPE (BBPE) tokenizer from scratch, with merge selection optimized by a **priority queue + doubly linked list** — no repeated full-corpus rescans
- Built a small **~63MB LLM from scratch in PyTorch**, covering RMSNorm, attention layers with **RoPE** positional encoding, and the full training stack
- Completed pre-training on the **CultureX zh dataset**, independently owning the full pipeline from corpus preprocessing and tokenization to training

cleanrl Reproduction – RL Algorithms from Scratch

- Reimplemented **DQN, TRPO, PPO, and SAC** from scratch following the cleanrl framework, achieving stable convergence on Gymnasium tasks (Lunar Lander, Taxi, CliffWalking, Mountain Car)
- Mastered algorithm internals — target networks, GAE, clipped objectives, twin critics — validating implementations against reference results

Open Source Contributions

urdf-visualizer (219 GitHub stars)

- Fixed xacro syntax compatibility issues, enabling the plugin to correctly parse xacro-formatted URDF files.

Competitions

ByteDance AI Full-Stack Challenge – Viral Video Style Transfer Track (1 of 12 tracks)

- Built an Agent system to analyze viral video features and automatically generate video clips with the Remotion framework. **Awarded direct interview qualification**